

BC 10. Which of the following functions are continuous at $x = 1$?

1993

SC

I. $\ln x$

II. e^x

III. $\ln(e^x - 1)$

- (A) I only (B) II only (C) I and II only (D) II and III only (E) I, II, and III

BC 79. Let f be the function given by $f(x) = \frac{(x-1)(x^2-4)}{x^2-a}$. For what positive values of a is f continuous for all real numbers x ?

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- (A) None
(B) 1 only
(C) 2 only
(D) 4 only
(E) 1 and 4 only

BC 5. Let f be the function defined by the following.

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$$f(x) = \begin{cases} \sin x, & x < 0 \\ x^2, & 0 \leq x < 1 \\ 2-x, & 1 \leq x < 2 \\ x-3, & x \geq 2 \end{cases}$$

For what values of x is f NOT continuous?

- (A) 0 only (B) 1 only (C) 2 only (D) 0 and 2 only (E) 0, 1, and 2

BC 35. If k is a positive integer, then $\lim_{x \rightarrow +\infty} \frac{x^k}{e^x}$ is

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- (A) 0 (B) 1 (C) e (D) $k!$ (E) nonexistent